

ATMOS INSIGHT

A High-Precision Machine Learning & Explainable AI (XAI) System for Real-Time Air Quality Monitoring, 24-Hour Recursive Forecasting, and 7-Day Trend Prediction

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Certificate of Approval

This is to certify that the major project report entitled "**ATMOS INSIGHT**" submitted by the Department of Computer Science & Engineering is an authentic record of work carried out under supervision.

Project Supervisor

Abstract

Severe urban air pollution across the Indo-Gangetic Plain poses critical public health hazards, requiring highly accurate, real-time, and explainable air quality forecasting systems. This project introduces Atmos Insight, an end-to-end, multi-tier artificial intelligence platform designed for real-time Air Quality Index (AQI) estimation, 24-hour recursive step-forward forecasting, and 7-day daily trend prediction.

Atmos Insight leverages a hybrid machine learning pipeline trained on multi-year CPCB monitoring telemetry from Kanpur, India, and live OpenWeatherMap satellite/station feeds. The core predictive engine comprises three specialized models: (1) a Random Forest Regressor for instant Current AQI estimation achieving an R^2 of 0.9997; (2) an XGBoost Recursive Forecaster for 24-hour hourly prediction achieving an R^2 of 0.9798; and (3) an Optuna-tuned XGBoost Direct Forecaster for 7-day daily forecasting achieving an R^2 of 0.2041.

To address model opacity, the platform embeds TreeSHAP to provide global feature importance rankings and local instance-level force/waterfall plots. The system is deployed via a FastAPI asynchronous backend integrated with a React 19 + Tailwind CSS v4 dashboard.

Chapter 1: Introduction

1.1 Background & Context

Urbanization, rapid industrial expansion, vehicular emissions, and seasonal agricultural biomass burning have transformed the Indo-Gangetic Plain into one of the most atmospheric-polluted regions globally.

1.2 Objectives

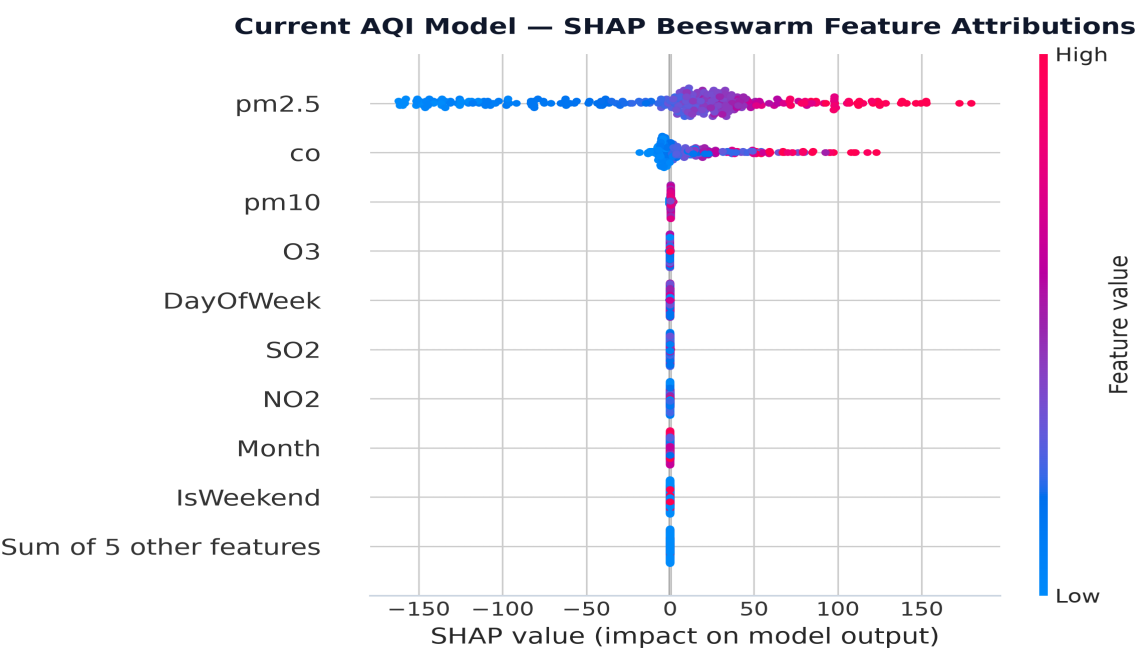
- Develop multi-tier machine learning models for instantaneous, 24-hour, and 7-day AQI prediction.
- Integrate TreeSHAP explainability for model transparency.
- Build a high-performance FastAPI backend and React 19 dashboard.

Chapter 8: Experimental Results

Model Pipeline	Algorithm	R ² Score	MAE	RMSE	MAPE (%)
Model 1: Current AQI	Random Forest	0.9997	0.9409	1.4809	0.72%
Model 2: 24-Hour Forecast	XGBoost	0.9798	2.3525	11.9653	1.63%
Model 3: 7-Day Forecast	XGBoost	0.2041	36.6951	49.1304	15.41%

Visualizations & Explainability

Figure: SHAP Beeswarm Summary Plot



Caption: Global SHAP summary plot displaying non-linear feature attribution across criteria pollutants.

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